LAB 03

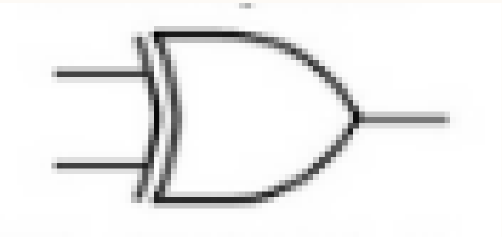
LOGIC GATES & ADDERS

CONTENTS

- Behavior of XOR Logic Gate
- Basic Logic gate implementation of adder

EXCLUSIVE OR – XOR LOGIC GATE

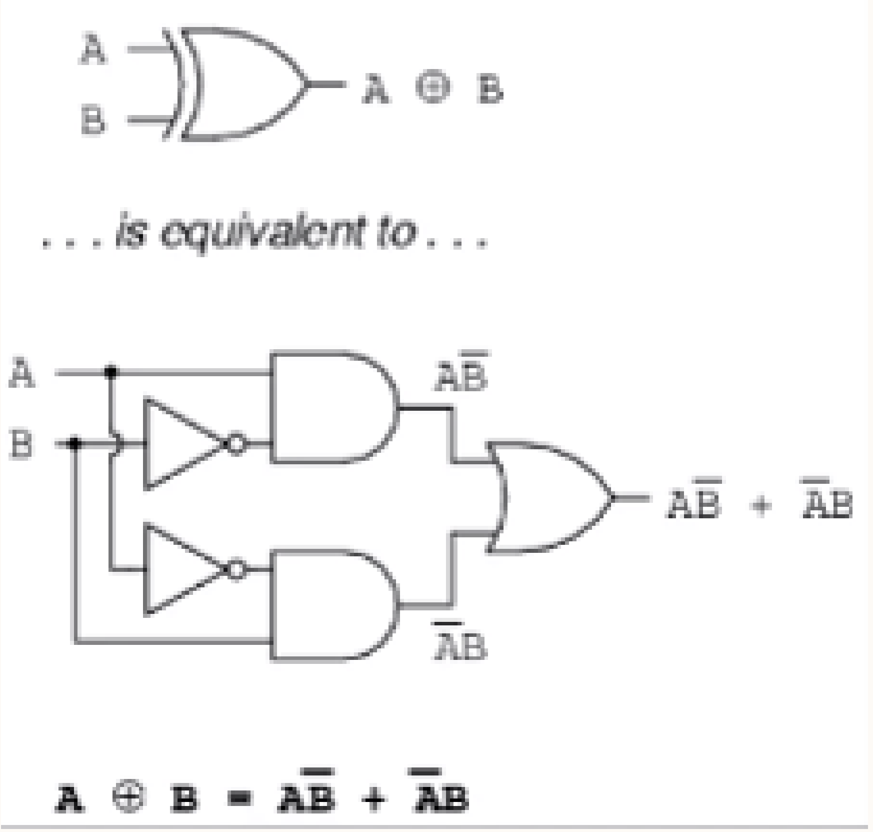
- XOR gate is a digital logic gate that give a true (1 or HIGH) output when the number of true inputs is odd.
- Boolean Expression for an XOR gate is add sign in circle ‘⊕’.

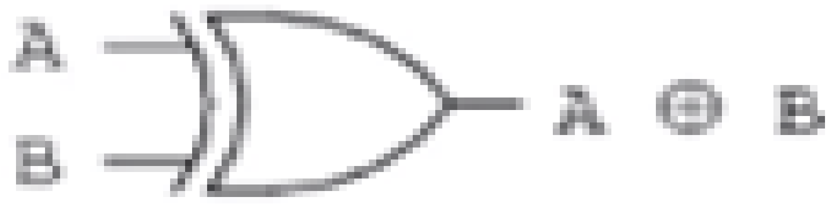


Logic Symbol

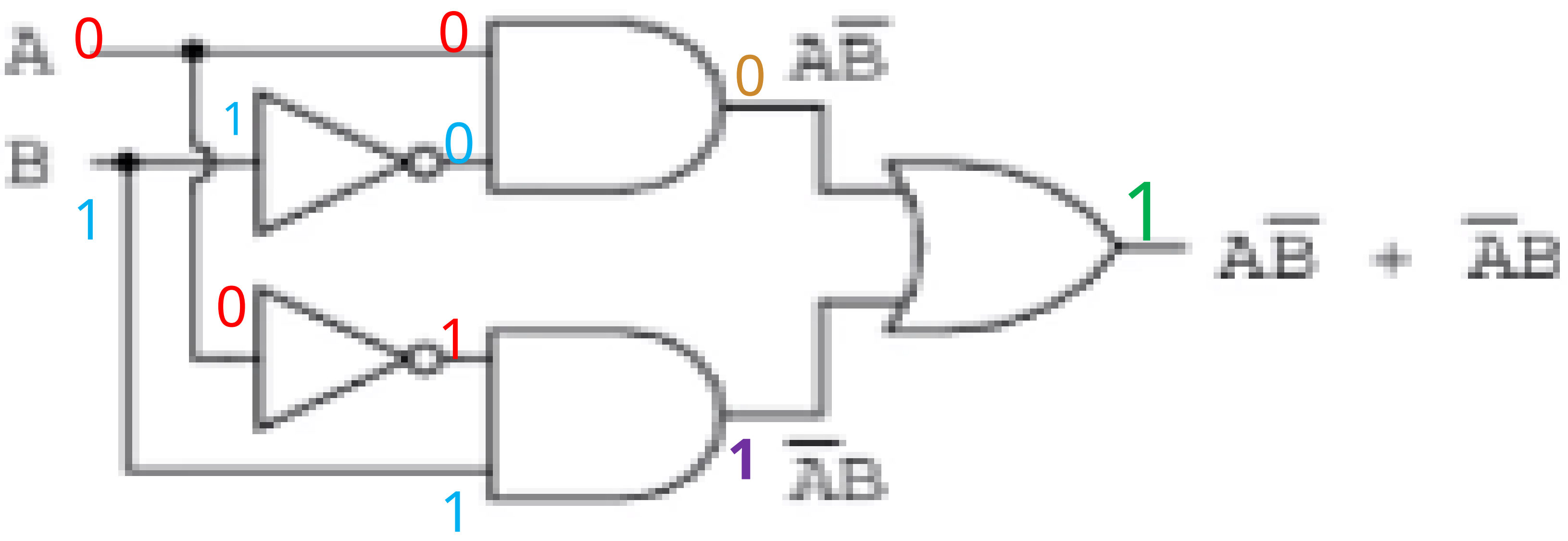
Input (A)	Input (B)	Output (Y)
0	0	0
0	1	1
1	0	1
1	1	0

$$Y = A \oplus B$$





. . . *is equivalent to* . . .



$$A \oplus B = A\overline{B} + \overline{A}B$$

Input (A)	Input (B)	Output (Y)
0	0	0
0	1	1
1	0	1
1	1	0

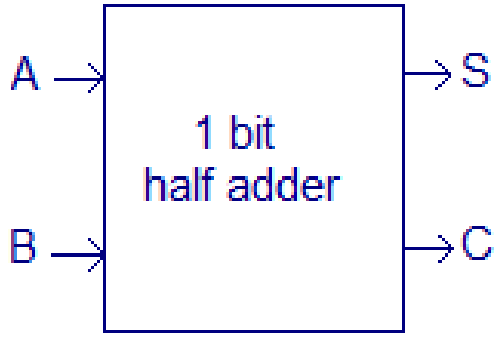
BASIC GATE LEVEL IMPLEMENTATION OF ADDER

HALF-ADDER

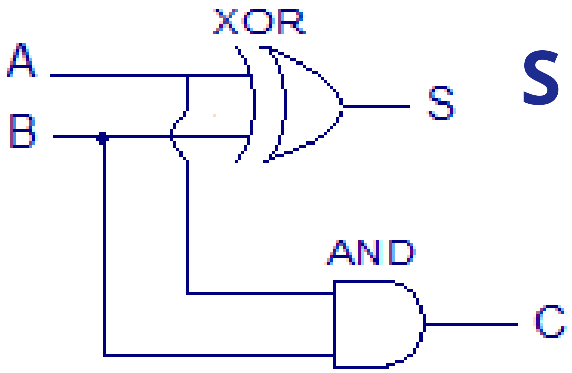
- *Half-Adder* is a combinational arithmetic circuit that adds two binary bits (A & B) and produces a sum bit (S) and carry bit (C) as the outputs.
- A half-adder circuit can be easily constructed using one XOR gate and one AND gate.

Inputs		Outputs	
A	B	S	C
0	0	0	0
1	0	1	0
0	1	1	0
1	1	0	1

Truth table



Schematic



Realization

$$S = A \oplus B$$

$$C = A \cdot B$$

IN- LAB TASKS

- **Task a:** Get familiarized with internal gate level structure of XOR gate IC 7486. (Uploaded on LMS Canvas or search it on web browser)
- **Task b:**
  1. Build logic implementation of two-inputs XOR gate – fill Table 3.2 ***(get it checked by RA)***
- 2. Build logic implementation of three-inputs XOR gate – fill Table 3.3 ***(get it checked by RA)***
- **Task c:**
  1. Build logic implementation of two-inputs half adder circuit – *left side of board* – fill Table 3.5 ***(get it checked by RA)***
- 2. Build logic implementation of two-inputs half adder circuit – *right side of board* – fill Table 3.6

- *Circuit should be neat and compact.
- *Do not remove the components from breadboard after *Task c*, **save it for LAB 04** - continue working on the same circuit